

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness — Enabling LLM Agents to Autonomously
Regulate Knowledge Utilization Through Situational Self-Awareness

arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Existing approaches blindly inject gold trajectories, external feedback, or domain knowledge regardless of whether the agent actually needs them at a given step.

Trajectory Imitation

e.g., ReAct

- Imitates gold trajectories at every step
- No situational awareness — always acts the same way
- Fragile on unexpected inputs; pattern collapse

Trial-and-Error Refinement

e.g., Reflexion, ETO

- Iterates with external feedback regardless of need
- Wastes compute on steps that were already correct
- Reflection even when direct action would suffice

Knowledge-Augmented Planning

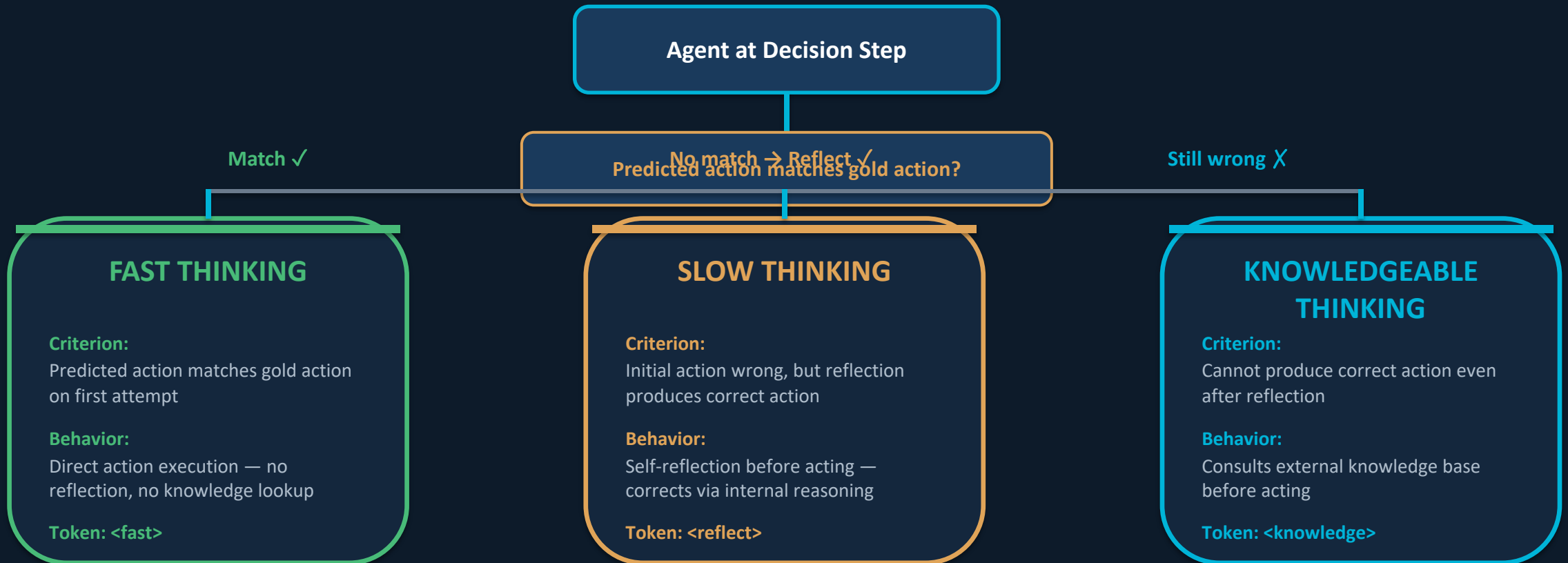
e.g., KnowAgent, WKM

- Injects domain knowledge at every single step
- 100% knowledge queries — maximum cost
- Knowledge can mislead when not actually needed

Humans possess situational self-awareness — the metacognitive ability to assess whether they can act directly, need to reflect, or must seek external help.

Three Situational Thinking Modes

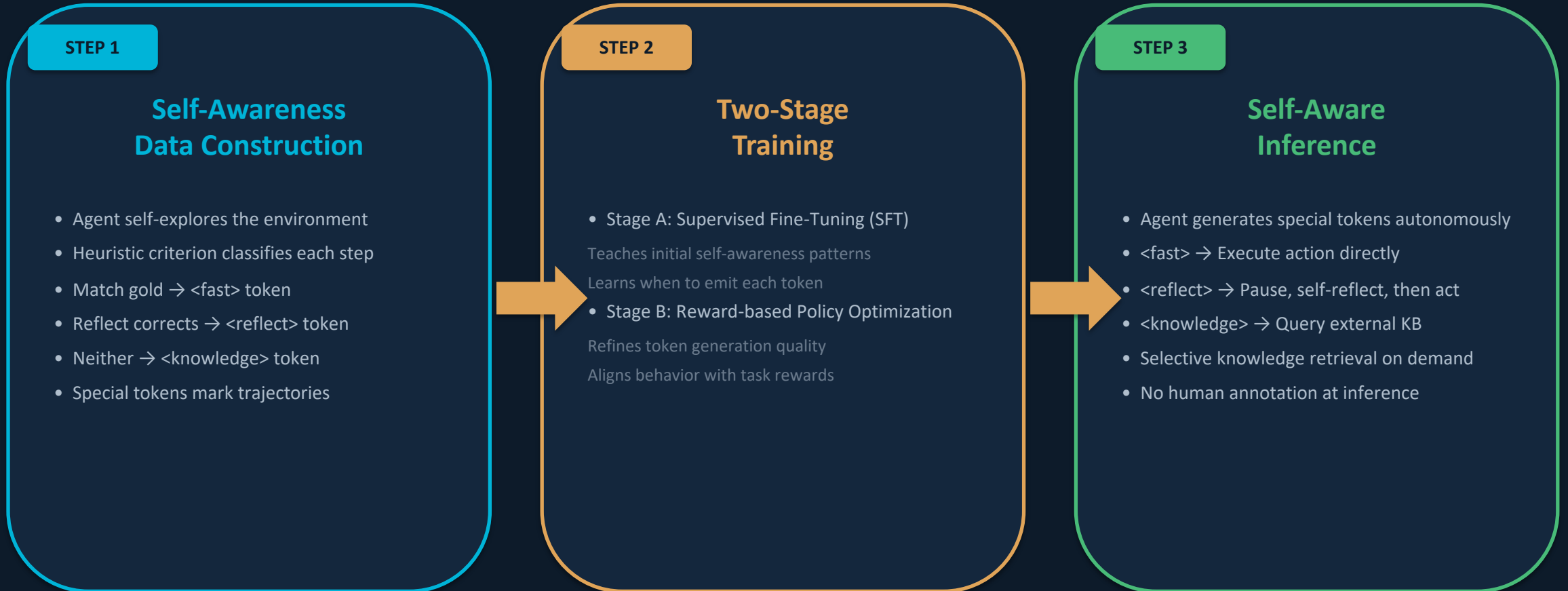
KnowSelf classifies each decision step into one of three cognitive situations



Analogous to human metacognition: we act instinctively when confident, pause to reflect when uncertain, and seek help when stuck.

KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference



Lightweight & data-efficient: requires only heuristic labeling on self-explored trajectories — no expensive human annotation or massive external feedback.

Qualitative Examples: KnowSelf vs. Baselines

KnowSelf correctly heats a mug and places a saltshaker where baselines fail

Task: "Put a hot mug in cabinet"

OpenAI-O1 (fails):

- Opens the microwave
- Incorrectly tries to take an egg from it
- No situational awareness that the mug needs heating

KnowSelf — Llama-8B (succeeds):

- Generates <reflect> token at the critical step
- Reflection: realizes the mug must be heated first
- Uses the open microwave to heat the mug
- Then places the hot mug in the cabinet ✓

Key Insight: Slow thinking (reflection) enables recognizing a hidden subgoal that direct action misses.

Task: "Put a saltshaker in drawer"

DeepSeek-R1 (fails):

- Observes the drawer is open
- Reflects that it should close it first (fails)
- Gets stuck in a loop — reflection without knowledge

KnowSelf — Llama-8B (succeeds):

- Generates <knowledge> token at the critical step
- Knowledge: agent should place the object directly, not remove unrelated items first
- Successfully places saltshaker in the drawer ✓

Key Insight: Knowledgeable thinking provides external guidance that reflection alone cannot discover.

ALFWorld Results: KnowSelf Achieves 84.33% with Llama-8B

Using only 15% knowledge — outperforms methods using 100% knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

KnowSelf rows highlighted in green | 100% knowledge methods shown in red | All = overall success rate (%)

Minimal Knowledge, Maximum Impact

KnowSelf's efficiency advantage over full-knowledge methods



15%

Knowledge usage
(Llama-8B)

84.33%

Overall accuracy
(Llama-8B)

26%

Knowledge usage
(Gemma-2B)

79.85%

Overall accuracy
(Gemma-2B)

vs

100%

Knowledge used by
ExpeL / KnowAgent / WKM

Selective knowledge use consistently outperforms indiscriminate injection — even smaller models benefit from situational self-awareness.

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection is wasteful and suboptimal

Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf's selective approach.

2

Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling

No expensive human annotation or massive external feedback — just self-explored trajectories with heuristic marks.

3

KnowSelf (Llama-8B) outperforms GPT-4o baselines: 84.33% vs 76.12% (ReAct) and 79.85% (ExpeL)

An 8B open-source model surpasses proprietary GPT-4o methods through self-aware knowledge regulation.

4

Only 15–26% knowledge usage needed versus 100% for competing methods

4–7× less knowledge queries while achieving higher accuracy — dramatic inference cost reduction.

5

The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL

Situational self-awareness transfers to smaller models, closing the gap with much larger systems

Future Directions:

Extending to multi-agent settings • Richer knowledge sources • Transfer across task domains • Adaptive situation thresholds